

DESIGN REFERENCE 01 | DUCT DESIGN QUICK REFERENCE

Airflow, velocity, area, friction rate & duct-sizing fundamentals

AIRFLOW	VELOCITY	AREA	ROUND AREA
CFM = Area (ft ²) × FPM	FPM = CFM / Area (ft ²)	Area (ft ²) = CFM / FPM	$A = \pi D^2 / 4$ Use D in feet

ROUND DUCT AIRFLOW QUICK TABLE

Dia.	Area	500 FPM	750 FPM	1,000 FPM	1,500 FPM	2,000 FPM
6"	0.20	98	147	196	295	393
8"	0.35	175	262	349	524	698
10"	0.55	273	409	545	818	1,091
12"	0.79	393	589	785	1,178	1,571
14"	1.07	535	802	1,069	1,604	2,138
16"	1.40	698	1,047	1,396	2,094	2,793
18"	1.77	884	1,325	1,767	2,651	3,534
20"	2.18	1,091	1,636	2,182	3,272	4,363
22"	2.64	1,320	1,980	2,640	3,960	5,280
24"	3.14	1,571	2,356	3,142	4,712	6,283
26"	3.69	1,844	2,765	3,687	5,531	7,374
28"	4.28	2,138	3,207	4,276	6,414	8,552
30"	4.91	2,454	3,682	4,909	7,363	9,817
32"	5.59	2,793	4,189	5,585	8,378	11,170
36"	7.07	3,534	5,301	7,069	10,603	14,137

Airflow values are calculated directly from $Q = A \times V$ and rounded to the nearest CFM. They are not recommended capacities; acceptable velocity depends on noise, pressure, application, and project criteria.

FRICTION RATE

Units: in. w.g. per 100 ft of duct.

ASHRAE's current duct-design guidance describes equal-friction sizing for downstream low-pressure systems using a design friction rate in the range of **0.05 to 0.20 in. w.g./100 ft**, while also limiting duct velocity for the application.

Important: friction rate is a design input - not a universal target. Verify project specifications, acoustical criteria, material/roughness, fittings, and available fan pressure.

RECTANGULAR DUCT

Area: $A = W \times H$ **Aspect ratio:** $AR = \text{long side} / \text{short side}$

Equivalent round diameter, common equal friction relation:

$$De \approx 1.30 \times (a \cdot b)^{0.625} / (a+b)^{0.25}$$

Use consistent dimensions for a and b. Equivalent diameter compares friction; it is not an area-equivalent diameter.

PRESSURE BASICS

Total Pressure (TP) = Static Pressure (SP) + Velocity Pressure (VP)

For standard air near 0.075 lb/ft³:

VP ≈ (FPM / 4005)² in. w.g.

Example: 1,500 FPM → VP ≈ 0.14 in. w.g.

DESIGN CHECK

1. Establish required CFM.
2. Select sizing method / design friction rate.
3. Size duct.
4. Check velocity and acoustics.
5. Calculate fitting + straight-duct losses.
6. Identify the critical path.
7. Verify fan external static pressure and project requirements.

REFERENCE VS. REQUIREMENT

Formula = mathematical relationship

Reference = published engineering guidance

Typical range = common design practice, not code

Project requirement = drawings/specifications/code/AHJ govern

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VELOCITY: USE CONTEXT, NOT A SINGLE 'MAGIC NUMBER'

Duct velocity is constrained by acoustics, pressure loss, space, system type, and project requirements. ASHRAE notes that lower velocity can reduce operating cost and duct-generated noise. Published limits are application-specific, so AnyHVAC does not label one velocity as universally 'recommended.'

Published example / context	Velocity / friction criterion	How to use it
ASHRAE example: duct above suspended acoustical ceiling; adjoining space NC/RC 35	Example cited in ASHRAE guidance: 3,000 FPM round; 1,750 FPM rectangular	Acoustical design example - not a universal limit
VA 2024: medium/high-pressure supply duct	Max 2,500 FPM and max 0.20 in. w.g./100 ft	Example owner design criteria
VA 2024: transfer-air duct	Max 750 FPM and max 0.04 in. w.g./100 ft	Example owner design criteria

These rows intentionally show why project context matters. Do not transfer an owner-specific criterion to another project without verification.

USEFUL CONVERSIONS

Quantity	IP relationship	SI relationship / note
Area	1 ft ² = 144 in ²	1 ft ² = 0.092903 m ²
Airflow	1 CFM	≈ 0.47195 L/s
Velocity	1 FPM	= 0.00508 m/s
Pressure	1 in. w.g.	≈ 249.1 Pa
Friction rate	0.10 in. w.g./100 ft	≈ 0.817 Pa/m

FAST SANITY CHECKS

Check	Question to ask
Airflow	Does the section CFM match the downstream load / terminal totals?
Velocity	Is the velocity appropriate for the acoustical and application criteria?
Friction	Does the selected size satisfy the design friction-rate approach?
Aspect ratio	Is the rectangular shape practical, efficient, and constructible?
Fittings	Have elbows, transitions, dampers, coils, filters, terminals, etc. been included?
Critical path	Is fan pressure based on the highest-loss path rather than a random branch?
Final verification	Do project specs, codes, manufacturer data, and AHJ requirements agree?



NEED THE EXACT CALCULATION?

AnyHVAC Duct Calculator
www.anyhvac.net/tools/duct-calculator
 Free. No sign-up required.

PRIMARY TECHNICAL REFERENCES

ASHRAE Handbook - Fundamentals, 2025, Ch. 21 Duct Design (current online IP chapter). U.S. Department of Veterans Affairs HVAC Design Manual, March 1, 2024. Equations and airflow-table values on this sheet are independently calculated by AnyHVAC.
Engineering Notice: AnyHVAC calculators and reference materials are informational and design-assistance tools. Results and criteria must be independently verified before final design, construction, equipment selection, permitting, or other safety-critical use.